

K. J. Somaiya College of Engineering, Mumbai-77
(Autonomous College Affiliated to University of Mumbai)

End Semester Exam
PARTIAL AUTO KT (April) 2016

Max. Marks: 100

Class: SE

Name of the Course: Applied Mathematics-IV

Course Code: U17C401 / U17EC401

Duration: 3 Hrs

Semester: IV

Branch: I.T / COMP

MUCBGS

Instructions:

(1) All Questions are Compulsory

(2) Figures to right indicate full marks. Each sub-question has equal marks.

Question NO.	Questions	Marks
Q.1		20
	A Attempt any TWO	12
	(i) If $A = \begin{bmatrix} 2 & 3 \\ -3 & -4 \end{bmatrix}$ Find A^{-1} & A^{50} using Characteristic equations	
	(ii) If $A = \begin{bmatrix} \frac{1+i}{2} & \frac{-1+i}{2} \\ \frac{1+i}{2} & \frac{1-i}{2} \end{bmatrix}$ then show that Characteristic roots are of unit modulus	
	(iii) Show that $A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$ derogatory and find its minimal polynomial.	
	B Attempt any ONE	8
	(i) State Cayley-Hamilton & verify it for $A = \begin{bmatrix} 0 & c & -b \\ -c & 0 & a \\ b & -a & 0 \end{bmatrix}$ find A^{-1} .	
Q.2	(ii) If $A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$ find eigen values & eigen vectors of A^2	
		20
	A Attempt any TWO	12
	(i) Expand $f(z) = \frac{2-z^2}{z(z^2-3z+2)}$ using the Laurent's series for $ z < 2$ & $1 < z $	
	(ii) Evaluate $\int_c \frac{(z^2)}{(z^4-1)} dz$ where (i) $c: z-1 = 1$ (ii) $ z = \frac{1}{4}$ (iii) $ z+i = 1$	
	(iii) State Cauchy's residue theorem & hence evaluate $\int_0^{2\pi} \frac{d\theta}{5+3\sin\theta}$	
	B Attempt any TWO	8
	(i) Evaluate $\int_c \frac{1-e^{2z}}{(z^3)} dz$ where c is the circle $ z = 1$	

	(ii)	Evaluate $\int_c \frac{1}{z^3(z+4)} dz$ where $c: z = 2$																			
	(iii)	Evaluate $\int_0^{1+i} (x - y + ix^2) dz$ along the path $x = y^2$																			
Q.3			20																		
	A	Attempt any TWO	12																		
	(i)	Player A and B toss a fair die alternately. One who gets a six first wins Rs 44, A starts. Find their mathematical expectations.																			
	(ii)	The income of a group of 10,000 persons was found to be normally distributed with mean of Rs. 750 & S.D. of Rs 50. What is lowest income of richest 250?																			
	(iii)	A manufacturer who produces medicine bottles finds out that 0.1% of the bottles are defective. Each box containing 500 bottles and he sells 100 boxes. Find how many boxes will contain no defective and at least one defective bottles.																			
	B	Attempt any TWO	8																		
	(i)	If distribution function $F(x)$ is defined as $F(x) = 1 - e^{-\frac{x}{4}}$, $x \geq 0$, then Find $p(4 < x < 8)$, $p(4 < x)$																			
(ii)	A random variable X follows Poisson distribution such that $P(X=1) = 2P(X=2)$. Find the mean and variance of the distribution. Also find $P(X=3)$.																				
(iii)	A random variable X has the following probability distribution <table><tr><td>X</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td></tr><tr><td>p(x)</td><td>K</td><td>2k</td><td>3k</td><td>k²</td><td>k² + k</td><td>2k²</td><td>4k²</td></tr></table> (i) Find the constant k. (ii) Find mean & Variance	X	1	2	3	4	5	6	7	p(x)	K	2k	3k	k ²	k ² + k	2k ²	4k ²				
X	1	2	3	4	5	6	7														
p(x)	K	2k	3k	k ²	k ² + k	2k ²	4k ²														
Q.4			20																		
	A	Attempt any TWO	12																		
		If $6y+x=6$ & $3x+2y=10$ are two regression lines, find coefficient of correlation. Also estimate y when $x=12$																			
	(ii)	A sample of 400 male students is found to have mean height of 67.47 inches. Can it be reasonably regarded as sample from a large population with mean 67.39 inches and standard deviation 1.30 inches?																			
	(iii)	A drug was administered to 5 persons and the systolic blood pressure before and after was measured. The result are given below <table><tr><td>Candidates</td><td>I</td><td>II</td><td>III</td><td>IV</td><td>V</td></tr><tr><td>B.P. before</td><td>140</td><td>130</td><td>132</td><td>150</td><td>140</td></tr><tr><td>B.P. after</td><td>132</td><td>126</td><td>133</td><td>144</td><td>133</td></tr></table> Test whether the drug is effective in lowering the systolic blood pressure. (Take 1% level of significance)	Candidates	I	II	III	IV	V	B.P. before	140	130	132	150	140	B.P. after	132	126	133	144	133	
	Candidates	I	II	III	IV	V															
	B.P. before	140	130	132	150	140															
B.P. after	132	126	133	144	133																

	B	Attempt any ONE										8																				
	(i)	Find Karl Pearson's coefficient of correlation & hence obtain (i) equation of regression lines of x on y (ii) equation of regression lines of y on x																														
	<table><tr><td>X</td><td>78</td><td>36</td><td>98</td><td>25</td><td>75</td><td>82</td><td>90</td><td>62</td><td>65</td><td>39</td></tr><tr><td>Y</td><td>84</td><td>51</td><td>91</td><td>60</td><td>68</td><td>62</td><td>86</td><td>58</td><td>53</td><td>47</td></tr></table>											X	78	36	98	25	75	82	90	62	65	39	Y	84	51	91	60	68	62	86	58	53
X	78	36	98	25	75	82	90	62	65	39																						
Y	84	51	91	60	68	62	86	58	53	47																						
(ii)	Investigate the association between the darkness of eye colour in mother & son for the following data																															
	<table><tr><td colspan="2" rowspan="2"></td><td colspan="3">Colour of mother's eyes</td></tr><tr><td>Dark</td><td>Not dark</td><td>Total</td></tr><tr><td rowspan="3">Colour of son's eyes</td><td>Dark</td><td>48</td><td>90</td><td>138</td></tr><tr><td>Notdark</td><td>80</td><td>782</td><td>862</td></tr><tr><td>Total</td><td>128</td><td>872</td><td>1000</td></tr></table>													Colour of mother's eyes			Dark	Not dark	Total	Colour of son's eyes	Dark	48	90	138	Notdark	80	782	862	Total	128	872	1000
		Colour of mother's eyes																														
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	Total	128	872	1000																												
.5												20																				
	A	Attempt any TWO										12																				
	(i)	Solve the following L.P.P. by Simplex method maximise $z = 6x_1 - 2x_2 + 3x_3$ Subject to $2x_1 - x_2 + 2x_3 \leq 2$ $x_1 + 4x_3 \leq 4$ $x_1, x_2, x_3 \geq 0$																														
	(ii)	Using the method of Lagrange's Multiplier solve the following N.L.P.P. Maximize $z = 6x_1 + 8x_2 - x_1^2 - x_2^2$ Subject to $4x_1 + 3x_2 = 16$ $3x_1 + 5x_2 = 15$ where $x_1, x_2 \geq 0$																														
	(iii)	Using the big-M method, solve the following L.P.P. Minimise $z = 10x_1 + 3x_2$ Subject to $x_1 + 2x_2 \geq 3$ $x_1 + 4x_2 \geq 4$ where $x_1, x_2 \geq 0$																														
B	Attempt any ONE(Mathematical Programming)										8																					
(i)	Using the Kuhn-Tucker conditions, solve the following N.L.P.P. Maximise $z = x_1^2 + x_2^2$ Subject to $x_1 + x_2 \leq 4$ $2x_1 + x_2 \leq 5$ where $x_1, x_2 \geq 0$																															
(ii)	Use the dual simplex method to solve the following L.P.P. maximise $z = 2x_1 + 2x_2 + 4x_3$ Subject to $2x_1 + 3x_2 + 5x_3 \geq 2$ $3x_1 + x_2 + 7x_3 \leq 3$ $x_1 + 4x_2 + 6x_3 \leq 5$ where $x_1, x_2, x_3 \geq 0$																															